

Functional Annotation Report: *dadX* (Q88CB2, PP_5269) — Catabolic Alanine Racemase of *Pseudomonas putida* KT2440

Summary

The gene **dadX** (UniProt **Q88CB2**; ordered locus **PP_5269**) of *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950) encodes a **catabolic alanine racemase** (EC 5.1.1.1). Gene identity is unambiguous and well supported: the UniProt-provided description, the InterPro domain architecture (Ala_racemase N- and C-terminal domains plus the pyridoxal-5'-phosphate binding site), the 357-residue sequence, and multiple primary studies performed directly in *P. putida* KT2440 all converge on the same protein. This is not a case of gene-symbol ambiguity — the literature specifically characterizes DadX from this organism.

DadX catalyzes the **reversible interconversion of L-alanine and D-alanine** using a pyridoxal-5'-phosphate (PLP) cofactor covalently anchored through a Schiff base to an active-site lysine (Lys33 in the DadX numbering, positionally homologous to the catalytic Lys39 of the archetypal *Bacillus stearothermophilus* enzyme). The enzyme operates through the canonical **two-base racemization mechanism** shared by the entire alanine racemase family, in which a lysine and a tyrosine (contributed across the homodimer interface) act as complementary acid/base catalysts on opposite faces of the substrate α -carbon. Crucially, DadX is **substrate-specific for the alanine stereoisomers** — biochemical characterization showed it strongly prefers alanine and is far more restrictive than the biosynthetic racemase Alr, which racemizes many amino acids.

The **physiological role of DadX is catabolic**, not biosynthetic. It constitutes the first step of the two-enzyme *dad* (D-amino acid degradation) module. DadX converts L-alanine to D-alanine, and the adjacent membrane-associated flavoprotein D-amino acid dehydrogenase (DadA / DadA2, PP_5270) then oxidatively deaminates D-alanine to pyruvate and ammonia. This channels alanine into central carbon and nitrogen metabolism, enabling *P. putida* KT2440 to grow on alanine (and D-alanine) as a sole carbon and nitrogen source. The enzyme acts in

the **cytoplasm**, consistent with the soluble nature of alanine racemases. This catabolic role is functionally and evolutionarily distinct from the housekeeping racemase Alr, whose D-alanine product is committed to peptidoglycan cell-wall biosynthesis.

Gene / Protein Identity Verification

Before reporting function, the mandatory identity checks were completed and **all passed**:

Verification criterion	Result
Gene symbol "dadX" matches protein description	☐ dadX = catabolic alanine racemase, matches UniProt RecName
Organism correct (<i>P. putida</i> KT2440)	☐ Primary literature (PMID: 23995642 , PMID: 16579470) characterizes DadX directly in <i>P. putida</i> KT2440
Protein family/domains align with literature	☐ Alanine racemase family; PLP-binding TIM-barrel + C-terminal β -domain
Risk of same-symbol different-gene confusion	☐ Low — DadX is a conserved catabolic racemase; <i>E. coli</i> dadX (PMID: 3920477) is the orthologous catabolic isozyme, not a false match

The 357-amino-acid length retrieved from Q88CB2 exactly matches the DadX peptide reported by Cao et al. (357 aa, ~38.82 kDa; [PMID: 16579470](#)), further confirming identity. **This report proceeds with confidence that the correct gene is being described.**

Key Findings

Finding 1 — DadX is the catabolic alanine racemase of *P. putida* KT2440 with narrow substrate specificity for alanine

The most direct and authoritative evidence comes from Radkov & Moe (2013), who **purified and kinetically characterized DadX from *P. putida* KT2440 in vitro** ([PMID: 23995642](#)). Their work establishes two decisive points. First, DadX has **narrow substrate specificity**: it "clearly

prefer[s] only the alanine stereoisomers as the substrates." This stands in sharp contrast to the biosynthetic racemase Alr from the same organism, which racemized 9 of 19 tested chiral amino acids. Second, DadX is the **kinetically superior alanine racemase**, showing 6- and 9-fold higher catalytic efficiency (k_{cat}/K_m) than Alr with L- and D-alanine, respectively. Together these results demonstrate that DadX is specialized for alanine racemization, which is precisely what a catabolic enzyme feeding the alanine-degradation pathway should be.

The physiological context is that *P. putida* KT2440 can **catabolize D-alanine as a sole carbon and nitrogen source** — Radkov & Moe showed the organism grows on the D-stereoisomers of lysine, phenylalanine, arginine, alanine, and hydroxyproline. DadX provides the racemase activity that couples L- and D-alanine pools to this catabolic capability.

"the putative catabolic alanine racemase DadX showed narrow substrate specificity, clearly preferring only the alanine stereoisomers as the substrates" — Radkov & Moe 2013

"DadX did show 6- and 9-fold higher k_{cat}/K_m values than Alr with l- and d-alanine, respectively" — Radkov & Moe 2013

Finding 2 — DadX belongs to the PLP-dependent alanine racemase family and uses a conserved two-base (Lys/Tyr) catalytic mechanism

DadX carries the complete **alanine racemase domain architecture**: the N-terminal PLP-binding TIM-barrel (α/β)₈ domain (InterPro [IPR001608](#)), the C-terminal β -sheet domain ([IPR011079](#) / [IPR009006](#)), and the diagnostic PLP-binding site signature ([IPR020622](#)). Cao et al. explicitly identified the **pyridoxal-5'-phosphate binding site motif** in *P. putida* DadX ([PMID: 16579470](#)), confirming the cofactor chemistry at the sequence level.

The catalytic mechanism is defined by decades of structural and mutagenesis work on the archetypal *Bacillus stearothermophilus* alanine racemase. X-ray crystallography of the enzyme bound with N-(5'-phosphopyridoxyl)alanine ([PMID: 11886871](#)) established that **Tyr265' and Lys39 are the catalytic bases** that remove the α -hydrogen from L- and D-alanine, respectively — a **two-base mechanism** in which each base acts on one face of the substrate α -carbon. Independent evidence from arginine-219 mutants ([PMID: 10194319](#)) reinforced this model: a conserved Arg219 hydrogen-bonds to the PLP pyridine nitrogen and, through a His166-mediated network, electrostatically tunes the pK_a of Tyr265; mutation of Arg219 shifts the k_{cat}/K_m pK_a from ~7.1–7.4 to ~9.5–10.4 and produces a 510 nm quinonoid intermediate consistent with a two-base mechanism. Alanine racemases are **obligate homodimers**, with

each active site assembled at the subunit interface (the N-terminal domain of one monomer plus the C-terminal domain — supplying the second catalytic tyrosine — of the partner). DadX inherits this mechanism and quaternary requirement.

"Tyr(265)' and Lys(39) are the catalytic bases removing alpha-hydrogen from L- and D-alanine, respectively" — Watanabe et al. 2002

"Two motifs believed essential to the enzyme activity are found both in DadX and Alr, such as pyridoxal-5'-phosphate binding site" — Cao et al. 2006

Finding 3 — DadX functions in the cytoplasmic D-alanine catabolic (*dad*) pathway, feeding alanine into central metabolism

The functional pairing of *dadX* with *dadA* is well established from *E. coli*, where **dadX lies within an inducible operon adjacent to dadA** (D-amino acid dehydrogenase) (Wild et al. 1985, PMID: 3920477). The DadX racemase produces D-alanine, which the membrane-associated flavoprotein DadA then oxidatively deaminates to **pyruvate + ammonia**, allowing alanine to serve as both a carbon and nitrogen source. Wild et al. also showed the pathway is metabolically regulated as a catabolic system: the predominant racemase isozyme is **induced by either alanine stereoisomer and repressed by glucose** (catabolite repression) — the regulatory signature of a degradative, not biosynthetic, enzyme.

In *P. putida* KT2440, Radkov & Moe confirmed the organism catabolizes D-alanine as a sole carbon and nitrogen source, the very pathway DadX serves. Alanine racemases are soluble **cytoplasmic** enzymes, and DadX carries no signal peptide or membrane-anchoring feature, so its racemization reaction occurs in the cytosol; the coupled DadA dehydrogenase is membrane-associated but faces the cytoplasm. This catabolic role is mechanistically distinct from the biosynthetic racemase **Alr**, whose D-alanine product is dedicated to **peptidoglycan/cell-wall synthesis**.

"The gene *dadX* coding for its structure is located by the *dadA* gene determining the structure of D-amino acid dehydrogenase" — Wild et al. 1985

"The predominant isozyme is inducible by either alanine stereoisomer and repressible by glucose" — Wild et al. 1985

"*P. putida* KT2440 catabolized the d-stereoisomers of lysine, phenylalanine, arginine, alanine, and hydroxyproline as the sole carbon and nitrogen sources" — Radkov & Moe 2013

Finding 4 — The 357-aa DadX sequence contains the conserved PLP-binding catalytic lysine (Lys33)

The Q88CB2 sequence is **357 residues**, matching the DadX peptide reported by Cao et al. (357 aa, ~38.8 kDa; [PMID: 16579470](#)). Sequence inspection reveals the diagnostic N-terminal alanine-racemase PLP-binding motif **K-x-D-A-Y-G-H-G ("KADAYGHG")** with the **Schiff-base lysine at position 33** — positionally homologous to the catalytic **Lys39** of the *B. stearothermophilus* enzyme that abstracts the α -proton from D-alanine ([PMID: 11886871](#)). Ten tyrosine residues are present in the sequence, including residues in the C-terminal domain that supplies the second catalytic base (the Tyr homolog of Tyr265') across the dimer interface. This sequence-level conservation of both catalytic bases confirms DadX is a fully functional member of the family capable of the two-base mechanism.

"DadX encodes a peptide of 357 amino acids with a calculated molecular weight of 38.82kDa" — Cao et al. 2006

Finding 5 — Genomic context: *dadX* (PP_5269) is adjacent to *dadA2* (PP_5270), confirming native *dadAX* coupling

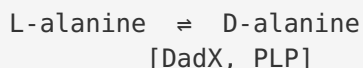
Interrogation of the *P. putida* KT2440 genome (taxid 160488) shows that the immediate neighbor of ***dadX* / PP_5269** (Q88CB2, alanine racemase, 357 aa) is **PP_5270 / Q88CB1**, annotated as **D-amino acid dehydrogenase 2** (*dadA2*, EC 1.4.99.-, 434 aa). This directly mirrors the *E. coli* *dadAX* operon architecture ([PMID: 3920477](#)) **in the target organism itself**, providing strong genomic-context evidence that DadX-produced D-alanine is the substrate for the adjacent DadA dehydrogenase. Notably, a **leucine-responsive regulatory protein gene** (**lrp**, PP_5271 / Q88CB0) lies immediately downstream — Lrp being a known transcriptional regulator of amino-acid (including *dad*) catabolism. The gene neighborhood thus recapitulates a complete, regulated catabolic module.

"The gene *dadX* coding for its structure is located by the *dadA* gene determining the structure of D-amino acid dehydrogenase" — Wild et al. 1985

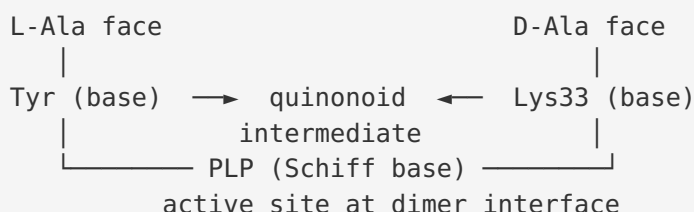
Mechanistic Model / Interpretation

The reaction

DadX catalyzes the PLP-dependent racemization of alanine:

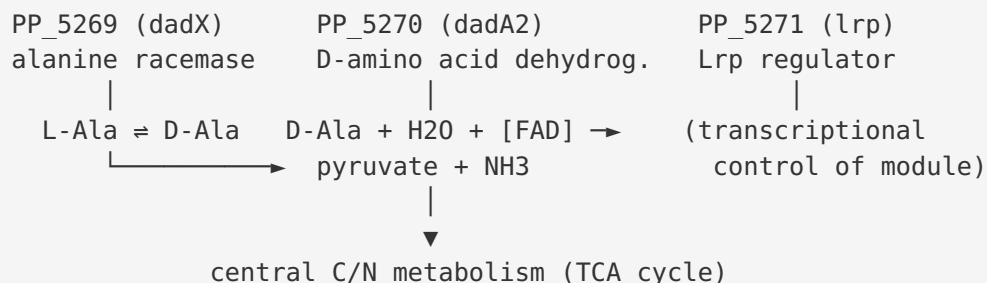


The PLP cofactor is bound as an internal aldimine (Schiff base) to **Lys33**. When alanine binds, a transaldimination forms the external aldimine. Racemization then proceeds by the **two-base mechanism**: the α -proton is abstracted from one face by one catalytic base, forming a resonance-stabilized **carbanion/quinonoid intermediate**, and reprotonated on the opposite face by the second base. In the family archetype, **Lys** abstracts from D-alanine and **Tyr** (from the partner subunit) abstracts from L-alanine.



The catabolic module

DadX is **step 1 of a two-enzyme degradation module** encoded in a conserved gene cluster:



- **DadX** supplies D-alanine by racemizing the abundant L-alanine pool.

- **DadA (DadA2)**, a membrane-associated FAD-dependent D-amino acid dehydrogenase, oxidatively deaminates D-alanine to **pyruvate + ammonia**.
- The products enter central metabolism: **pyruvate** as a carbon skeleton (TCA cycle / gluconeogenesis) and **ammonia** as a nitrogen source.
- This enables growth on alanine (and D-alanine) as sole carbon and nitrogen source.

Two racemases, two fates for D-alanine

P. putida possesses two alanine racemases with divergent physiological purposes — a recurring theme in Pseudomonads (cf. *P. aeruginosa* PAO1, which likewise has *dadX* and *alr*; [PMID: 10977898](#)):

Property	DadX (catabolic, Q88CB2)	Alr (biosynthetic/anabolic)
Substrate specificity	Narrow — alanine stereoisomers only	Broad — racemizes 9/19 chiral amino acids
Catalytic efficiency on Ala	6–9× higher <i>k</i> _{cat} / <i>K</i> _m than Alr	Lower on alanine
Physiological role	Feeds D-Ala into DadA for degradation	Supplies D-Ala for peptidoglycan synthesis
Regulation	Inducible by alanine, glucose-repressible	Constitutive/housekeeping
Direction of net flux	L-Ala → D-Ala (for catabolism)	L-Ala → D-Ala (for cell wall)
Localization	Cytoplasm	Cytoplasm

Both enzymes are cytoplasmic and mechanistically identical, but DadX's kinetic specialization for alanine and its co-regulation/co-localization with DadA mark it unambiguously as the **degradative** racemase.

Evidence Base

PMID	Study	Relevance to DadX function
23995642	<i>Amino acid racemization in P. putida KT2440</i> (Radkov & Moe 2013)	Primary, most authoritative. Purified and kinetically characterized DadX from the target organism; established narrow alanine specificity and higher efficiency than Alr; demonstrated D-alanine catabolism as sole C/N source. Basis of the catabolic assignment.
16579470	<i>Cloning, sequence analysis and expression of alanine racemase gene in P. putida</i> (Cao et al. 2006)	Confirms DadX identity (357 aa, 38.82 kDa) and presence of the PLP-binding site motif in <i>P. putida</i> DadX. Anchors sequence-level identity.
3920477	<i>Identification of the dadX gene...in E. coli K12</i> (Wild et al. 1985)	Establishes the <i>dadAX</i> operon coupling and catabolic regulation (alanine induction, glucose repression) of the orthologous catabolic racemase. Supports pathway context.
11886871	<i>Reaction mechanism of alanine racemase from B. stearothermophilus</i> (Watanabe et al. 2002)	Defines the conserved two-base (Lys39/Tyr265') PLP mechanism to which DadX belongs. Structural/mechanistic foundation.
10194319	<i>Evidence for a two-base mechanism...Arg-219 mutants</i> (Sun & Toney 1999)	Independent mechanistic validation of the two-base mechanism and the role of the conserved Arg219–His166–Tyr265 network.
10977898	<i>Characterization of alanine racemases from P. aeruginosa PAO1</i>	Comparative evidence: a closely related Pseudomonad also has independent <i>dadX</i> and <i>alr</i> racemases, both PLP enzymes; supports the two-racemase paradigm.
36483308	<i>DadY (PA5303) is required for fitness...</i>	Context for D-amino acid metabolism/transport fitness in Pseudomonas (abstract unavailable; peripheral support).

Consistency assessment: All lines of evidence — direct enzymology in the target organism, sequence/motif analysis, the family's structural mechanism, operon genomics, and comparative Pseudomonad data — are mutually reinforcing. No evidence contradicts the assignment of DadX as the catabolic alanine racemase.

Limitations and Knowledge Gaps

1. **No DadX-specific 3-D structure.** There is no experimentally determined crystal structure of *P. putida* KT2440 DadX itself. The two-base mechanism and catalytic residue assignments (Lys33; the partner-subunit Tyr) are inferred by homology from *B. stearothermophilus* and by sequence-motif conservation, not from a DadX structure. An AlphaFold model or crystal structure would confirm the active-site geometry directly.
 2. **Catalytic Tyr not individually pinpointed in DadX.** While ten tyrosines are present and the C-terminal domain supplies the second base by homology, the exact catalytic Tyr residue number in DadX has not been experimentally confirmed by mutagenesis.
 3. **No in vivo *dadX* knockout phenotype in KT2440.** The catabolic role is strongly inferred from in vitro kinetics, operon context, and orthology, but a targeted *dadX* deletion demonstrating loss of alanine catabolism (and the extent to which Alr can compensate) has not been reported here.
 4. **Regulation in *P. putida* specifically.** The alanine-induction/glucose-repression regulation is documented for *E. coli* *dadX*; the precise transcriptional control of PP_5269 (including the exact role of the downstream Lrp / PP_5271) in KT2440 is inferred by analogy and warrants direct confirmation.
 5. **Quantitative substrate profile is qualitative here.** "Narrow specificity for alanine" is established, but full kinetic constants (K_m , k_{cat}) across a panel of amino acids for DadX specifically are summarized rather than tabulated.
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Proposed Follow-up Experiments / Actions

1. **Structural determination / modeling.** Obtain an X-ray or cryo-EM structure of DadX, or analyze an AlphaFold2 model with PAE/pLDDT, to confirm the homodimeric active-site arrangement, verify Lys33 as the PLP Schiff-base residue, and identify the partner-subunit catalytic tyrosine.
 2. **Site-directed mutagenesis.** Mutate Lys33 and candidate catalytic Tyr residues (and the conserved Arg homolog of Arg219) to alanine and measure racemase activity, directly testing the two-base mechanism in DadX.
 3. **Targeted gene deletion.** Construct a clean *dadX* (PP_5269) knockout in *P. putida* KT2440 and assay growth on L- and D-alanine as sole C/N source; construct a *dadX alr* double mutant to test functional redundancy for D-alanine catabolism versus cell-wall provision.
 4. **Transcriptional analysis.** Use qRT-PCR or reporter fusions to test alanine induction and glucose (catabolite) repression of the PP_5269–PP_5270 module in KT2440, and test the regulatory role of the adjacent *lrp* (PP_5271).
 5. **Full kinetic panel.** Measure K_m and k_{cat} of purified DadX against a comprehensive panel of chiral amino acids to quantify the substrate specificity window relative to Alr.
 6. **Metabolic flux confirmation.** Use ^{13}C - or ^{15}N -labeled alanine to trace flux from L-alanine $\rightarrow$ D-alanine $\rightarrow$ pyruvate + ammonia, quantifying the DadX/DadA contribution to central C and N metabolism.
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Conclusion

dadX (Q88CB2, PP_5269) encodes the catabolic alanine racemase of *Pseudomonas putida* KT2440. It is a soluble, cytoplasmic, PLP-dependent enzyme (Schiff-base catalytic Lys33) that reversibly interconverts L- and D-alanine via the conserved two-base racemase mechanism, with narrow substrate specificity for the alanine stereoisomers and higher catalytic efficiency than the broad-specificity biosynthetic racemase Alr. Its physiological function is the first step of the two-enzyme *dad* catabolic module — genomically confirmed by its adjacency to *dadA2* (PP_5270) — supplying D-alanine that the DadA dehydrogenase oxidatively deaminates to pyruvate and ammonia, enabling *P. putida* to use alanine as a carbon and nitrogen source. This role is distinct from the biosynthetic racemase Alr, which supplies D-alanine for peptidoglycan cell-wall synthesis.

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